Qn. Set Code-7(SOT)

Semester: 4th
Programme: B.Tech
Branch: CSE, IT, CSCE, CSSE

SPRING END SEMESTER EXAMINATION-2025

4th Semester B.Tech

OPERATING SYSTEMS

CS20002 / IT20002 / CM20002 / CC20002 / CS2002

(For 2023 & Previous Admitted Batches)

Time: 2 Hours 30 Minutes | Full Marks: 50

Answer any FIVE questions.
Question paper consists of two SECTIONS i.e. A and B.
Section A is compulsory.
Attempt any Four question from Sections B.
The figures in the margin indicate full marks.
Candidates are required to give their answers in their own words as far as practicable and all parts of a question should be answered at one place only.

SECTION-A

1. Answer the following questions. [1 × 10]

(a) What is the concept of a safe state in deadlock avoidance?

(b) A disk has a seek time of 2 ms for the first 1 mm of movement, with each additional 1 mm taking 1.5 ms. If the disk head moves from track 10 to track 85, where each track is 0.5 mm apart, determine the total seek time.

(c) Explain how user-level threads differ from kernel-level threads and their impact on performance.

(d) In a dynamically allocated memory system, a process of size X MB needs to be executed within a total of Y MB of free memory, which is divided into n non-contiguous blocks (B1, B2, ..., Bn). How can external fragmentation be expressed mathematically in this scenario as the cause of allocation failure?

(e) Define and differentiate between starvation and deadlock.

(f) Differentiate between compile time and load time address binding.

(g) Given the free memory blocks (in KB): 120, 450, 320, 610, 250, 580 and the arrival of processes (in KB) in order: 230, 140, 490, 350. Allocate the processes using both Best Fit and Worst Fit strategies. Identify which strategy minimizes fragmentation.

(h) If page size is of 4KB, logical address is of 22bits and Page table entry is 4B, then find out the size of the page table.

(i) Implement a semaphore-based solution where a process P1 prints "Hello", then process P2 prints "World!" in the correct order.

(j) What are the advantages and disadvantages of Shortest Job First (SJF) compared to Round Robin (RR) and FCFS in terms of average waiting time and starvation?

SECTION-B

2. (a) A system with 5 frames processes the following page reference string: 7, 0, 1, 2, 0, 3, 4, 2, 3, 0, 3, 2, 1, 2, 0, 1, 7, 0, 1. [5]

   i. Compute number of page faults using FIFO and LRU algorithms.

   ii. Check for Belady's Anomaly in FIFO when increasing to 6 frames, with calculations.

   iii. Calculate the total penalty for each algorithm if each page fault incurs 5 ms.

   (b) Why does address translation in an inverted page table require a linear search or hashed search instead of direct indexing? How does this search-based translation impact performance compared to the direct indexing method used in a hierarchical page table? [5]

3. (a) Explain the different file allocation methods used by an operating system, highlighting their advantages and disadvantages. [5]

(b) A disk system consists of 500 cylinders, numbered from 0 to 499. The disk head is initially positioned at cylinder 250 and is moving towards higher-numbered cylinders. The request queue contains the following cylinder numbers: 100, 350, 230, 90, 420, 300, 370. The seek rate is 2 ms per cylinder. Determine the number of cylinder movements and the total seek time using SSTF, SCAN and LOOK scheduling algorithms. [5]

4. (a) What is the concept of a domain of protection in an operating system? How is it implemented using an Access Matrix? [5]

(b) In a single-level paging system, the virtual address space is 256 MB, and each page table entry (PTE) occupies 4 bytes. Determine the minimum page size required so that the entire page table fits within a single page. [5]

5. (a) State the necessary conditions that must be met to implement a solution for the critical section problem. Explain how Peterson's two-process solution satisfies these conditions. [5]

(b) In a Unix-like file system, an index node (inode) contains 12 direct pointers, one single-indirect pointer, and one double-indirect pointer. Given that the disk block size is 4 KB and each disk block address is 32 bits (4 bytes) long, determine the maximum possible file size that can be supported. [5]

6. (a) A multilevel queue scheduling system has three queues with the following scheduling policies: [5]

i. Real-time Processes (Highest Priority, SJF)

ii. System Processes (Priority Scheduling, lower value = higher priority)

iii. Interactive Processes (Round Robin, 5 ms time quantum)

Given the following processes:

Process	Queue Type	Arrival Time	Burst Time	Priority (for System Queue)
P1	Real-time	0	8	-
P2	System	1	5	2
P3	Interactive	2	8	-
P4	System	4	6	1
P5	Interactive	5	9	-

Draw the Gantt Chart showing execution order. Compute Average Waiting Time and Average Turnaround Time.

(b) Consider a system with 5 processes (P1, P2, P3, P4, P5) and 3 resource types (A, B, C). The allocation and maximum demand matrices for each process are provided below: [5]

Process	Max (A, B, C)	Allocation (A, B, C)
P1	(7, 5, 3)	(0, 1, 0)
P2	(3, 2, 2)	(2, 0, 0)
P3	(9, 0, 2)	(3, 0, 2)
P4	(4, 2, 2)	(2, 1, 1)
P5	(5, 3, 3)	(0, 0, 2)

The total available resources in the system are A = 10, B = 5, C = 7. Using the Banker's Algorithm, determine the following:

i. If a safe sequence exists, identify the sequence.

ii. If process P2 requests (1, 0, 2) resources, can the system allocate them safely?

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